

ग्राम व्यापार सहायक (Rural Advisory Assistant)

AI & Deterministic Decision Intelligence for Rural Micro-Entrepreneurs

Comprehensive Technology Stack & Engineering Architecture Specification

Status: Ready for Leadership Review
Version: v1.0.0 (Production Architecture)
Project: Smart India Hackathon (SIH)

Executive Technical Summary

The **Rural Advisory Assistant** is engineered as a zero-hallucination, dual-delivery advisory platform. It combines a modern **Next.js 14 + TypeScript** reactive web application with a **zero-dependency offline standalone engine**. The architecture enforces a strict **Three-Tier Separation Principle**: financial calculations and government policy eligibility are 100% deterministic and legally precise, while the AI layer is restricted exclusively to vernacular narration and voice synthesis.

Core 3-Way Architectural Separation

1. Deterministic Engine

Pure Mathematical Rigor: Reducing-balance EMI, compounding post-moratorium interest, DSCR, break-even, 1.5x Risk Buffer, & cash surplus. Zero AI hallucination for all numbers.

2. Policy & Rules Engine

Statutory Compliance: Automated scheme routing (MoSJE / Stand-Up India guidelines), health classification (Thriving/Surviving/At-Risk), & single-buyer concentration flags (>60%).

3. Vernacular AI Narrator

Plain-Language Explanation: Converts computed numbers into voice-ready Hindi, Hinglish, & English scripts with 100% computational attribution tags.

Technical Stack & Component Breakdown

LAYER / DOMAIN	TECHNOLOGIES & LIBRARIES	ENGINEERING CAPABILITIES & SCOPE
Frontend & Framework	Next.js 14.2 (App Router)React 18.2TypeScript 5.3	• Server & Client Component hybrid architecture for optimal performance • Strict static typing across business profiles, loan schemes, & health models • Dynamic client-side reactive computation & state handling
Design System & UI	Tailwind CSS 3.4Radix UI PrimitivesLucide IconsCVA	• Accessible UI primitives (dialogs, select dropdowns, tabs, progress bars) • Custom fintech & vernacular design tokens optimized for mobile displays
Financial & Math Engine	TypeScript EnginePure Functional Math	• Reducing-balance EMI & compounding post-moratorium interest formulas • 1.5x Risk Buffer calculation (mitigates "Silent Moratorium Killer") • Live DSCR (Debt Service Coverage Ratio) recomputation from actuals
Data Visualization	Recharts 2.10	• Interactive quarterly cash surplus & revenue trajectory curves • Loan repayment amortization & interest vs. principal distribution charts
Macro & Micro Data	Census ProjectionsNSSO Expenditure Data	• Hyper-local supply-demand proxy estimation across 8 enterprise categories • Block and village population density matrix across 10+ Indian states
Reporting & Export	NABARD DPR EnginePrint CSS Media	• Generates bank-ready Detailed Project Reports (DPR) • 100% compliant with NABARD Model Bankable Project formats
Field & Offline Delivery	Standalone Vanilla JSZero-Dependency HTML	• Single-file portable execution (app-standalone.html + engine.js) • Runs completely offline in remote rural environments without internet connectivity

Strategic Value Pillars for Leadership & Evaluators

1. 100% Trust & Zero Hallucinations

All credit metrics derived via audited formulas; LLMs never generate financial figures.

2. Voice-First & Vernacular

Tailored for low digital literacy with conversational Hinglish & Hindi audio scripts.

3. Bank-Grade DPR Export

Transforms informal micro-data into NABARD-compliant DPRs for bank credit approval.

4. Continuous Health Tracking

Ongoing quarterly monitoring (Thriving / Surviving / At-Risk) with early default warnings.